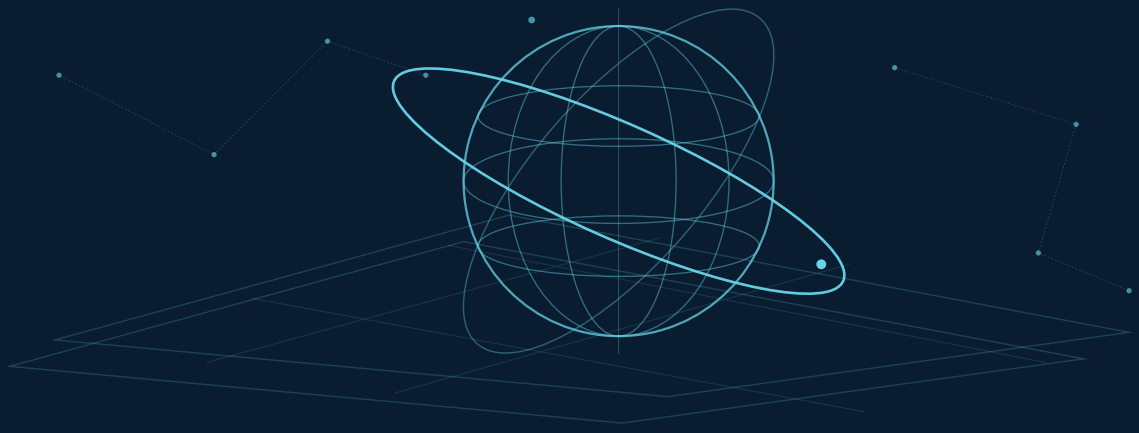


SPECIFICATION

Booking desk composition

Three use cases sharing an entity, a validation, a tool, and a quality requirement.

ATLAS-EX-COMPOSITION

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- composition
- booking
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Context, actors, and scope

A booking desk creates, locates, and cancels bookings. The example is a specification, not an application.

The scope covers these three cases. Payment, customer management, and calendar integration are outside the example.

This is a synthetic specification example. Source SRC-BOOKING-BRIEF is a teaching baseline, not evidence of stakeholder approval or a claimed Needs-to-L1 derivation.

ACTOR ACT-BOOKING-CLERK

Name

Booking clerk

Kind: person

Description

A person operating the booking desk.

TERM VOC-BOOKING

Term

Booking

Definition

A retained reservation with a code, title, and lifecycle status.

Synonyms: none

Entity and shared validations

The booking entity defines data, states, a transition, and two validations.

TYPE TYP-BOOKING-STATUS

Name

Booking status

Definition

The lifecycle state of a booking.

Constraints

Known values are active and cancelled.

Examples

- active
- cancelled

ENTITY ENT-BOOKING

Name

Booking

Meaning

A booking records a named reservation and has a lifecycle status.

Identity

Description

Two bookings are the same when their codes are equal.

Fields: FLD-BOOKING-CODE

Fields

FLD-BOOKING-CODE

Name

Booking code

Type: text

Cardinality: 1

Required: true

Description

A stable uppercase code that identifies the booking.

Constraints

1 to 12 uppercase letters or digits.

Examples

Valid

- BOOK001

Invalid

- book-001

FLD-BOOKING-TITLE

Name

Title

Type: text

Cardinality: 1

Required: true

Description

A short human-readable title.

Constraints

At least one non-whitespace character.

Examples

Valid

- Annual review

Invalid

-

FLD-BOOKING-STATUS

Name

Status

Type: booking-status

Cardinality: 1

Required: true

Description

The lifecycle state of the booking.

Constraints

One of active or cancelled.

Examples

Valid

- active
- cancelled

Invalid

- deleted

FLD-BOOKING-VERSION

Name

Version

Type: integer

Cardinality: 1

Required: true

Description

Concurrency token: zero for a new ready Booking, positive for a retained Booking.

Constraints

A non-negative integer; successful save increments it.

Examples

Valid

- 0
- 1

Invalid

- -1

Relationships: none

States

ST-BOOKING-ACTIVE

Name

Active

Description

The booking reserves its item.

ST-BOOKING-CANCELLED

Name

Cancelled

Description

The booking no longer reserves its item and is retained.

Transitions

TRN-BOOKING-CANCEL

Name

Cancel

From: ST-BOOKING-ACTIVE

To: ST-BOOKING-CANCELLED

Description

Cancellation changes status and retains the record.

Validations

VLD-BOOKING-CODE

Name

Booking code format

Scope: field

Statement

The code is valid when it has 1 to 12 characters, all uppercase letters or digits.

Inputs

Name

Booking code

Source: field

Meaning

The value supplied for FLD-BOOKING-CODE.

Conditions

C-VLD-BOOK-CODE-EMPTY

Condition

The code is non-empty.

Meaning

An empty code fails.

C-VLD-BOOK-CODE-LEN

Condition

The length is between 1 and 12.

Meaning

Longer codes fail.

C-VLD-BOOK-CODE-CHARS

Condition

Every character is A-Z or 0-9.

Meaning

Lowercase and punctuation fail.

Results

Outcome: pass

Description

The code is accepted unchanged.

Outcome: fail

Description

The code is rejected; the field keeps its previous value.

Flow: GRAPH-VLD-BOOKING-CODE

VLD-BOOKING-CANCEL

Name

Cancellation is allowed only from active

Scope: transition

Statement

A cancellation transition is valid only when the booking's current status is active.

Inputs

Name

Current status

Source: state

Meaning

The status read from the located booking.

Conditions

C-VLD-BOOK-CANCEL-ACTIVE

Condition

The current status is active.

Meaning

A cancelled booking fails this condition.

Results

Outcome: pass

Description

Cancellation may proceed.

Outcome: fail

Description

The already-cancelled refusal applies; no save occurs.

Flow: GRAPH-VLD-BOOKING-CANCEL

VLD-BOOKING-TITLE

Name

Booking title validity

Scope: field

Statement

The title is text with at least one non-whitespace character. Preserve valid text exactly.

Inputs

Name

Title

Source: field

Meaning

FLD-BOOKING-TITLE candidate.

Conditions

C-VLD-BOOK-TITLE-TEXT

Condition

The title has at least one non-whitespace character.

Meaning

Empty and whitespace-only titles fail.

Results

Outcome: pass

Description

Accept unchanged.

Outcome: fail

Description

Refuse before save.

Flow: GRAPH-VLD-BOOKING-TITLE

Examples

EX-BOOKING-VALID

Kind: valid

Description

An active booking with a conforming code.

Data

```
{
"code": "BOOK001",
"title": "Annual review",
"status": "active",
"version": 0
}
```

EX-BOOKING-INVALID

Kind: invalid

Description

A booking with a lowercase code and unknown status.

Data

```
{
"code": "book-001",
"title": "Annual review",
"status": "deleted",
"version": 0
}
```

Assumptions: none

Open issues: none

Shared abstract tool

One abstract tool serves all three cases through two operations.

TOOL TOL-BOOKING-STORE

Name

Booking store

Purpose

Load and retain bookings. The contract is shared by creation, lookup, and cancellation and does not select a database.

Operations

OP-BOOKING-LOAD

Name

Load booking

Purpose

Read one retained booking by code.

Inputs

Name

Booking code

Type: text

Meaning

The exact code to locate.

Required: true

Conditions

The store is available.

Inputs described as ready entities already satisfy canonical entity invariants. Do not duplicate those validations inside this tool. When constructing entities from raw retained records, use the canonical entity hydration rules; operation-specific external conditions remain this tool's responsibility.

Results

Name

Booking

Type: BOOKING

Meaning

The retained booking with its current status.

Guarantees

The result reflects the stored record at read time.

Effects

No persisted change.

Failures

Code: NOT_FOUND

Meaning

No retained booking has the code.

Consumer distinction

Distinct from an invalid code and from an unavailable store.

Code: STORE_UNAVAILABLE

Meaning

The store could not complete the read.

Consumer distinction

Must not be reported as not found.

Guarantees

A successful load returns one current booking.

OP-BOOKING-SAVE

Name

Save booking

Purpose

Persist the current data of one booking.

Inputs

Name

Booking

Type: BOOKING

Meaning

The booking to persist, including its status.

Required: true

Name

Expected version

Type: integer

Meaning

The version observed by the caller.

Required: true

Conditions

The store is available.

Inputs described as ready entities already satisfy canonical entity invariants. Do not duplicate those validations inside this tool. When constructing entities from raw retained records, use the canonical entity hydration rules; operation-specific external conditions remain this tool's responsibility.

Results

Name

Saved booking

Type: BOOKING

Meaning

The booking with an incremented version.

Guarantees

A successful save confirms the new data.

Effects

The retained booking is replaced with the supplied data.

Failures

Code: VERSION_CONFLICT

Meaning

The expected version is no longer current.

Consumer distinction

This operation made no write. The existing or concurrently updated record is not overwritten.

Code: STORE_UNAVAILABLE

Meaning

The store did not confirm the write.

Consumer distinction

Distinguish write_outcome=not_applied, when supported by a no-write guarantee, from unknown. Do not report success or automatically retry an unknown write.

Guarantees

A successful save confirms one updated retained booking.

Expected version zero means insert only if the identifier is absent. A positive expected version means replace only the matching retained version. Comparison and write are atomic; success increments the version by one.

VERSION_CONFLICT guarantees this operation made no change; another concurrent operation may have changed the stored record.

Use cases and composition

Cancellation calls lookup. The response requirement is defined once and referenced by cancellation.

USE CASE `UC-BOOKING-LOOKUP`

Name

Locate booking

Objective

Return the current data of one retained booking.

Actors

Actor: ACT-BOOKING-CLERK

Permission

Permitted to locate bookings.

Description

The clerk operates the booking desk.

Resolve this permission from trusted context for the concrete initiating caller; deny absent or unresolved evidence.

Trigger

A clerk requests a booking by code.

Context

The lookup is used directly by the desk and by cancellation.

Caller context must contain a resolved authenticated principal, the relevant case permissions, and a validity status. Its origin is the application's trusted identity-and-permission authority; request fields and caller-supplied role strings are not trusted authorization evidence. Anonymous execution is not permitted. Missing, unresolved, expired or insufficient context is denied before any validation, tool access or effect.

The initiating actor is ACT-BOOKING-CLERK. The actor is a role; the caller is the concrete principal. The identity/authentication mechanism and channel encoding remain later design decisions.

Inputs

Name

Booking code

Type: text

Meaning

The code to locate.

Required: true

Entry conditions

Requests may arrive without valid caller context; the explicit denial path handles them before any collaborator is invoked.

Entities

Entity: ENT-BOOKING

Validations: VLD-BOOKING-CODE

Purpose

Validate the code before loading.

Tools

Tool: TOL-BOOKING-STORE

Operation: OP-BOOKING-LOAD

Purpose

Load the retained booking.

Calls: none

Outcomes

OUT-BOOKING-LOOKUP-FOUND

Kind: success

Description

The booking is returned.

Effects

No persisted change.

OUT-BOOKING-LOOKUP-NOTFOUND

Kind: refusal

Description

No booking has the code.

Effects

No persisted change.

OUT-BOOKING-LOOKUP-FAILURE

Kind: failure

Description

The store is unavailable.

Effects

No persisted change.

OUT-BOOKING-LOOKUP-INVALID

Kind: refusal

Description

The booking code is invalid.

Effects

No load or data change.

OUT-BOOKING-LOOKUP-DENIED

Kind: refusal

Description

The trusted caller context does not authorize this case.

Effects

No validation, collaborator call, persisted change, or disclosure of booking existence.

Functional requirements

RF-BOOKING-LOOKUP-01

Statement

The case applies VLD-BOOKING-CODE and refuses an invalid code without loading.

Scope: case

Applies to: UC-BOOKING-LOOKUP

Flow references: N-BOOK-LOOK-CODE, N-BOOK-LOOK-CODEC, N-BOOK-LOOK-INVALID, E-BOOK-LOOK-3

RF-BOOKING-LOOKUP-02

Statement

The case calls OP-BOOKING-LOAD and returns the loaded booking when found.

Scope: operation

Applies to: UC-BOOKING-LOOKUP, OP-BOOKING-LOAD

Flow references: N-BOOK-LOOK-LOAD, N-BOOK-LOOK-FOUND, E-BOOK-LOOK-4, E-BOOK-LOOK-7

RF-BOOKING-LOOKUP-03

Statement

NOT_FOUND is reported as a distinct refusal with no effect.

Scope: operation

Applies to: UC-BOOKING-LOOKUP, OP-BOOKING-LOAD

Flow references: N-BOOK-LOOK-NOTFOUND, E-BOOK-LOOK-5

RF-BOOKING-LOOKUP-04

Statement

STORE_UNAVAILABLE is reported as a failure, not as not found.

Scope: operation

Applies to: UC-BOOKING-LOOKUP, OP-BOOKING-LOAD

Flow references: N-BOOK-LOOK-FAIL, E-BOOK-LOOK-6

RF-BOOKING-LOOKUP-AUTH

Statement

Require this case's permission in the trusted authenticated caller context. Deny missing, unresolved, expired or insufficient context before any validation, tool call, composed use-case call or effect. Anonymous execution is forbidden.

Scope: case

Applies to: UC-BOOKING-LOOKUP

Flow references: N-BOOKING-LOOKUP-AUTH, N-BOOKING-LOOKUP-DENIED, E-BOOKING-LOOKUP-ALLOW, E-BOOKING-LOOKUP-DENY

Nonfunctional requirements: none

Flows

Main: GRAPH-BOOKING-LOOKUP

Subflows: none

Flow text

Resolve and authorize the initiating caller first (RF-BOOKING-LOOKUP-AUTH). Denial terminates before any subsequent step.

Validate the code (RF-BOOKING-LOOKUP-01). An invalid code is refused before load.

Load through OP-BOOKING-LOAD (RF-BOOKING-LOOKUP-02). NOT_FOUND is a refusal (RF-BOOKING-LOOKUP-03); STORE_UNAVAILABLE is a failure (RF-BOOKING-LOOKUP-04).

Acceptance criteria

AC-BOOKING-LOOKUP-01

Statement

A retained code returns the booking and its current status.

Verifies: RF-BOOKING-LOOKUP-02

AC-BOOKING-LOOKUP-02

Statement

An unknown code returns not found, not an empty booking.

Verifies: RF-BOOKING-LOOKUP-03

AC-BOOKING-LOOKUP-03

Statement

An unavailable store is distinguished from not found.

Verifies: RF-BOOKING-LOOKUP-04

AC-BOOKING-LOOKUP-INVALID

Statement

A lowercase or punctuated code fails the canonical code rule and no load is invoked.

Verifies: RF-BOOKING-LOOKUP-01

AC-BOOKING-LOOKUP-AUTH

Statement

Missing, unresolved, expired or insufficient caller context produces denial with no validation, tool call, composed call or data disclosure.

Verifies: RF-BOOKING-LOOKUP-AUTH

Traceability

TRC-BOOKING-LOOKUP-01

Source: RF-BOOKING-LOOKUP-01

Targets: VLD-BOOKING-CODE

Note

Shared entity validation.

TRC-BOOKING-LOOKUP-ORIGIN

Source: SRC-BOOKING-BRIEF

Targets: RF-BOOKING-LOOKUP-01, RF-BOOKING-LOOKUP-02, RF-BOOKING-LOOKUP-03, RF-BOOKING-LOOKUP-04, RF-BOOKING-LOOKUP-AUTH

Note

Versioned synthetic origin for these example obligations.

Assumptions: none

Sources: SRC-BOOKING-BRIEF

Open issues: none

USE CASE UC-BOOKING-CREATE

Name

Create booking

Objective

Create one active booking and confirm its stored state.

Actors

Actor: ACT-BOOKING-CLERK

Permission

Permitted to create bookings.

Description

The clerk operates the booking desk.

Resolve this permission from trusted context for the concrete initiating caller; deny absent or unresolved evidence.

Trigger

A clerk supplies a code and title.

Context

The desk creates a booking in the shared store.

Caller context must contain a resolved authenticated principal, the relevant case permissions, and a validity status. Its origin is the application's trusted identity-and-permission authority; request fields and caller-supplied role strings are not trusted authorization evidence. Anonymous execution is not permitted. Missing, unresolved, expired or insufficient context is denied before any validation, tool access or effect.

The initiating actor is ACT-BOOKING-CLERK. The actor is a role; the caller is the concrete principal. The identity/authentication mechanism and channel encoding remain later design decisions.

Inputs

Name

Booking code

Type: text

Meaning

The code to create.

Required: true

Name

Title

Type: text

Meaning

The booking title.

Required: true

Entry conditions

Requests may arrive without valid caller context; the explicit denial path handles them before any collaborator is invoked.

Entities

Entity: ENT-BOOKING

Validations: VLD-BOOKING-CODE, VLD-BOOKING-TITLE

Purpose

Validate the code and form the active booking.

Tools

Tool: TOL-BOOKING-STORE

Operation: OP-BOOKING-SAVE

Purpose

Persist the new booking.

Calls: none

Outcomes

OUT-BOOKING-CREATE-DONE

Kind: success

Description

The active booking is created.

Effects

One new booking is retained.

OUT-BOOKING-CREATE-INVALID

Kind: refusal

Description

The code or title is invalid.

Effects

No save and no retained booking.

OUT-BOOKING-CREATE-CONFLICT

Kind: failure

Description

The store reports a version conflict or unavailability.

Effects

No created booking is reported.

OUT-BOOKING-CREATE-DENIED

Kind: refusal

Description

The trusted caller context does not authorize this case.

Effects

No validation, collaborator call, persisted change, or disclosure of booking existence.

Functional requirements

RF-BOOKING-CREATE-01

Statement

The case applies VLD-BOOKING-CODE and refuses an invalid code without saving.

Scope: case

Applies to: UC-BOOKING-CREATE

Flow references: N-BOOK-CREATE-CODE, N-BOOK-CREATE-CODEC, N-BOOK-CREATE-INVALID, E-BOOK-CREATE-3

RF-BOOKING-CREATE-02

Statement

The case creates an active booking and calls OP-BOOKING-SAVE with version 0.

Scope: operation

Applies to: UC-BOOKING-CREATE, OP-BOOKING-SAVE

Flow references: N-BOOK-CREATE-SAVE, E-BOOK-CREATE-4, E-BOOK-CREATE-7

RF-BOOKING-CREATE-03

Statement

VERSION_CONFLICT and STORE_UNAVAILABLE are reported as failures, never as a created booking.

Scope: operation

Applies to: UC-BOOKING-CREATE, OP-BOOKING-SAVE

Flow references: N-BOOK-CREATE-DUP, N-BOOK-CREATE-FAIL, E-BOOK-CREATE-5, E-BOOK-CREATE-6

RF-BOOKING-CREATE-04

Statement

A successful result returns the created active booking.

Scope: case

Applies to: UC-BOOKING-CREATE

Flow references: N-BOOK-CREATE-DONE, E-BOOK-CREATE-7

RF-BOOKING-CREATE-TITLE

Statement

After a valid code, apply VLD-BOOKING-TITLE. Refuse invalid title before creating a ready active Booking or invoking save.

Scope: case

Applies to: UC-BOOKING-CREATE

Flow references: N-BOOK-CREATE-TITLE, N-BOOK-CREATE-TITLEVALID, E-BOOK-CREATE-TITLE-2, E-BOOK-CREATE-TITLE-3

RF-BOOKING-CREATE-AUTH

Statement

Require this case's permission in the trusted authenticated caller context. Deny missing, unresolved, expired or insufficient context before any validation, tool call, composed use-case call or effect. Anonymous execution is forbidden.

Scope: case

Applies to: UC-BOOKING-CREATE

Flow references: N-BOOKING-CREATE-AUTH, N-BOOKING-CREATE-DENIED, E-BOOKING-CREATE-ALLOW, E-BOOKING-CREATE-DENY

Nonfunctional requirements

RNF-BOOKING-CREATE-RESPONSE

Scope: interaction

Applies to: UC-BOOKING-CREATE

Statement

The boundary presents the result of a creation within 2 seconds of the stimulus for inputs within the declared limits.

Verifiable criterion

Visible result time is at most 2 seconds.

Verification

Run 30 creation requests against the declared configuration and record elapsed time.

Metric

Name

Visible result time

Unit

seconds

Target: 2

Operator: <=

Measurement

Start

The creation stimulus is accepted.

End

The result is visible.

References: SRC-BOOKING-BRIEF

Applicability

Origin: SRC-BOOKING-BRIEF. This is a synthetic teaching obligation; any numeric target and scenario count are illustrative assumptions, not an accepted production SLA or a measurement made by this repository.

Flows

Main: GRAPH-BOOKING-CREATE

Subflows: none

Flow text

Resolve and authorize the initiating caller first (RF-BOOKING-CREATE-AUTH). Denial terminates before any subsequent step.

Validate the code with VLD-BOOKING-CODE (RF-BOOKING-CREATE-01).

Apply title validity after code validity; refuse invalid title without saving (RF-BOOKING-CREATE-TITLE).

Create the active booking and save through OP-BOOKING-SAVE (RF-BOOKING-CREATE-02).

A save failure is reported as failure (RF-BOOKING-CREATE-03); a success returns the created booking (RF-BOOKING-CREATE-04).

Acceptance criteria

AC-BOOKING-CREATE-01

Statement

A valid code creates an active booking and a later lookup returns it.

Verifies: RF-BOOKING-CREATE-02, RF-BOOKING-CREATE-04

AC-BOOKING-CREATE-02

Statement

An invalid code is refused with no retained booking.

Verifies: RF-BOOKING-CREATE-01

AC-BOOKING-CREATE-03

Statement

A version conflict is never presented as a created booking.

Verifies: RF-BOOKING-CREATE-03

AC-BOOKING-CREATE-TITLE

Statement

A whitespace-only title is refused and OP-BOOKING-SAVE is not invoked.

Verifies: RF-BOOKING-CREATE-TITLE

AC-BOOKING-CREATE-AUTH

Statement

Missing, unresolved, expired or insufficient caller context produces denial with no validation, tool call, composed call or data disclosure.

Verifies: RF-BOOKING-CREATE-AUTH

AC-BOOKING-CREATE-QUALITY

Statement

In the declared didactic configuration, demonstrate the canonical quality criterion and its stated verification procedure; preserve its status as a teaching assumption, not a measured production claim.

Verifies: RNF-BOOKING-CREATE-RESPONSE

Traceability

TRC-BOOKING-CREATE-01

Source: RF-BOOKING-CREATE-01

Targets: VLD-BOOKING-CODE

Note

Shared validation also used by lookup.

TRC-BOOKING-CREATE-02

Source: RF-BOOKING-CREATE-02

Targets: OP-BOOKING-SAVE

Note

Shared tool operation also used by cancellation.

TRC-BOOKING-CREATE-ORIGIN

Source: SRC-BOOKING-BRIEF

Targets: RF-BOOKING-CREATE-01, RF-BOOKING-CREATE-02, RF-BOOKING-CREATE-03, RF-BOOKING-CREATE-04, RF-BOOKING-CREATE-TITLE, RF-BOOKING-CREATE-AUTH, RNF-BOOKING-CREATE-RESPONSE

Note

Versioned synthetic origin for these example obligations.

Assumptions: none

Sources: SRC-BOOKING-BRIEF

Open issues: none

USE CASE UC-BOOKING-CANCEL

Name

Cancel booking

Objective

Retain the booking and change its status to cancelled when cancellation is allowed.

Actors

Actor: ACT-BOOKING-CLERK

Permission

Permitted to cancel bookings.

Description

The clerk operates the booking desk.

Resolve this permission from trusted context for the concrete initiating caller; deny absent or unresolved evidence.

Trigger

A clerk requests cancellation by code.

Context

Cancellation reuses lookup and the shared store.

Caller context must contain a resolved authenticated principal, the relevant case permissions, and a validity status. Its origin is the application's trusted identity-and-permission authority; request fields and caller-supplied role strings are not trusted authorization evidence. Anonymous execution is not permitted. Missing, unresolved, expired or insufficient context is denied before any validation, tool access or effect.

The initiating actor is ACT-BOOKING-CLERK. The actor is a role; the caller is the concrete principal. The identity/authentication mechanism and channel encoding remain later design decisions.

Inputs

Name

Booking code

Type: text

Meaning

The booking to cancel.

Required: true

Entry conditions

Requests may arrive without valid caller context; the explicit denial path handles them before any collaborator is invoked.

Entities

Entity: ENT-BOOKING

Validations: VLD-BOOKING-CANCEL

Purpose

Check that the located booking may move from active to cancelled.

Tools

Tool: TOL-BOOKING-STORE

Operation: OP-BOOKING-SAVE

Purpose

Persist the cancelled status.

Calls

Use case: UC-BOOKING-LOOKUP

Contract

Receives the located booking or the lookup refusal/failure.

Authorization

The cancellation actor must also be permitted to locate bookings.

Preserve the concrete principal and trusted caller context in the lookup invocation. Lookup independently checks its own permission; cancellation permission never bypasses it.

Outcomes

`OUT-BOOKING-CANCEL-DONE`

Kind: success

Description

The booking is cancelled and retained.

Effects

The booking status becomes cancelled.

OUT-BOOKING-CANCEL-ALREADY

Kind: refusal

Description

The booking is already cancelled.

Effects

No save and no data change.

OUT-BOOKING-CANCEL-NOTFOUND

Kind: refusal

Description

The lookup did not find the booking.

Effects

No save and no data change.

OUT-BOOKING-CANCEL-FAILURE

Kind: failure

Description

The store or lookup failed.

Effects

No cancellation is claimed.

OUT-BOOKING-CANCEL-INVALID

Kind: refusal

Description

Lookup refused an invalid code.

Effects

No save or data change.

OUT-BOOKING-CANCEL-DENIED

Kind: refusal

Description

The trusted caller context does not authorize this case.

Effects

No validation, collaborator call, persisted change, or disclosure of booking existence.

Functional requirements

RF-BOOKING-CANCEL-01

Statement

The case calls UC-BOOKING-LOOKUP and preserves its not-found and invalid-code refusals.

Preserve the caller context and propagate lookup denial before any save.

Scope: case

Applies to: UC-BOOKING-CANCEL

Flow references: N-BOOK-CANCEL-LOOK, N-BOOK-CANCEL-FOUND, N-BOOK-CANCEL-NOTFOUND, N-BOOK-CANCEL-INVALID, N-BOOK-CANCEL-LOOKDENIED, E-BOOK-CANCEL-3, E-BOOK-CANCEL-4, E-BOOK-CANCEL-INVALID, E-BOOK-CANCEL-LOOKDENIED, N-BOOK-REFUSE-NOTFOUND, E-BOOK-REFUSE-2

RF-BOOKING-CANCEL-02

Statement

The case applies VLD-BOOKING-CANCEL to the located booking before any save.

Scope: case

Applies to: UC-BOOKING-CANCEL

Flow references: N-BOOK-CANCEL-VALIDATE, N-BOOK-CANCEL-VALID, E-BOOK-CANCEL-5

RF-BOOKING-CANCEL-03

Statement

An already-cancelled booking produces the already-cancelled refusal with no save and no effect.

Scope: case

Applies to: UC-BOOKING-CANCEL

Flow references: N-BOOK-CANCEL-ALREADY, E-BOOK-CANCEL-6, N-BOOK-REFUSE-START, N-BOOK-REFUSE-ALREADY, E-BOOK-REFUSE-1

RF-BOOKING-CANCEL-04

Statement

A passing transition sets status cancelled and calls OP-BOOKING-SAVE with the observed version.

Scope: operation

Applies to: UC-BOOKING-CANCEL, OP-BOOKING-SAVE

Flow references: N-BOOK-CANCEL-SAVE, N-BOOK-CANCEL-DONE, E-BOOK-CANCEL-7, E-BOOK-CANCEL-9

RF-BOOKING-CANCEL-05

Statement

A store failure is reported distinctly and does not claim cancellation.

Scope: operation

Applies to: UC-BOOKING-CANCEL, OP-BOOKING-SAVE

Flow references: N-BOOK-CANCEL-ERROR, E-BOOK-CANCEL-8, E-BOOK-CANCEL-LOOKERROR

RF-BOOKING-CANCEL-AUTH

Statement

Require this case's permission in the trusted authenticated caller context. Deny missing, unresolved, expired or insufficient context before any validation, tool call, composed use-case call or effect. Anonymous execution is forbidden.

Scope: case

Applies to: UC-BOOKING-CANCEL

Flow references: N-BOOKING-CANCEL-AUTH, N-BOOKING-CANCEL-DENIED, E-BOOKING-CANCEL-ALLOW, E-BOOKING-CANCEL-DENY

Nonfunctional requirements

RNF-BOOKING-CANCEL-RESPONSE

Reference to: RNF-BOOKING-CREATE-RESPONSE

Scope: interaction

Applies to: UC-BOOKING-CANCEL

References: SRC-BOOKING-BRIEF

Applicability

Origin: SRC-BOOKING-BRIEF. This is a synthetic teaching obligation; any numeric target and scenario count are illustrative assumptions, not an accepted production SLA or a measurement made by this repository.

Flows

Main: GRAPH-BOOKING-CANCEL

Subflows: GRAPH-BOOKING-CANCEL-REFUSAL

Flow text

Resolve and authorize the initiating caller first (RF-BOOKING-CANCEL-AUTH). Denial terminates before any subsequent step.

Call UC-BOOKING-LOOKUP (RF-BOOKING-CANCEL-01). Its refusals and failures remain observable.

Apply VLD-BOOKING-CANCEL to the located booking (RF-BOOKING-CANCEL-02).

An already-cancelled booking reaches the refusal subflow (RF-BOOKING-CANCEL-03) with no save.

On pass, set status cancelled and save with the observed version (RF-BOOKING-CANCEL-04). A save failure is reported (RF-BOOKING-CANCEL-05).

A composed lookup denial terminates cancellation before save. A lookup failure is propagated as failure, never collapsed into not found.

Acceptance criteria

AC-BOOKING-CANCEL-01

Statement

Cancelling an active booking returns cancelled and retains the code.

Verifies: RF-BOOKING-CANCEL-02, RF-BOOKING-CANCEL-04

AC-BOOKING-CANCEL-02

Statement

Cancelling an already-cancelled booking is refused with no data change.

Verifies: RF-BOOKING-CANCEL-03

AC-BOOKING-CANCEL-03

Statement

A lookup not-found is distinguishable from an already-cancelled refusal.

Verifies: RF-BOOKING-CANCEL-01, RF-BOOKING-CANCEL-03

AC-BOOKING-CANCEL-FAILURE

Statement

An unavailable lookup or save remains a failure; an unknown write is not reported as no change, success, or retried automatically.

Verifies: RF-BOOKING-CANCEL-05

AC-BOOKING-CANCEL-CONTEXT

Statement

A caller allowed to cancel but not to locate is denied by the composed lookup, with the same principal preserved and no save.

Verifies: RF-BOOKING-CANCEL-01

AC-BOOKING-CANCEL-AUTH

Statement

Missing, unresolved, expired or insufficient caller context produces denial with no validation, tool call, composed call or data disclosure.

Verifies: RF-BOOKING-CANCEL-AUTH

AC-BOOKING-CANCEL-QUALITY

Statement

In the declared didactic configuration, demonstrate the canonical quality criterion and its stated verification procedure; preserve its status as a teaching assumption, not a measured production claim.

Verifies: RNF-BOOKING-CANCEL-RESPONSE

Traceability

TRC-BOOKING-CANCEL-01

Source: RF-BOOKING-CANCEL-02

Targets: VLD-BOOKING-CANCEL, TRN-BOOKING-CANCEL

Note

Transition validation and entity transition.

TRC-BOOKING-CANCEL-02

Source: RNF-BOOKING-CANCEL-RESPONSE

Targets: RNF-BOOKING-CREATE-RESPONSE

Note

Explicit shared obligation reference.

TRC-BOOKING-CANCEL-ORIGIN

Source: SRC-BOOKING-BRIEF

Targets: RF-BOOKING-CANCEL-01, RF-BOOKING-CANCEL-02, RF-BOOKING-CANCEL-03, RF-BOOKING-CANCEL-04, RF-BOOKING-CANCEL-05, RF-BOOKING-CANCEL-AUTH, RNF-BOOKING-CANCEL-RESPONSE

Note

Versioned synthetic origin for these example obligations.

Assumptions: none

Sources: SRC-BOOKING-BRIEF

Open issues: none

Interaction boundary

The command line exposes all three cases and maps their observable outcomes.

BOUNDARY BND-BOOKING-CLI

Name

Booking desk command line

Modality: cli

Purpose

Expose booking creation, lookup, and cancellation through commands.

Origin classification: this channel is an illustrative solution proposal. Its form, command syntax, layout and encoding are outside the L1 conformance claim. The referenced cases define domain inputs and outcomes independently of this channel.

Audience

Desk clerks operating a local terminal.

Interactions

INT-BOOKING-CREATE

Name

Create command

Description

Accepts a code and title and presents the created booking or a refusal/failure.

Illustration only; no channel-specific UI or command choice is promoted to a domain RF by appearing here.

Use case: UC-BOOKING-CREATE

Trigger

The clerk invokes booking create.

Input mapping

Source: argv.code

Target: UC-BOOKING-CREATE.input.code

Meaning

The booking code argument.

Source: argv.title

Target: UC-BOOKING-CREATE.input.title

Meaning

The title argument.

Result mapping

Source: OUT-BOOKING-CREATE-DONE

Target: stdout.created

Meaning

The active booking is created.

Source: OUT-BOOKING-CREATE-INVALID

Target: stderr.invalid

Meaning

The code or title is invalid.

Source: OUT-BOOKING-CREATE-CONFLICT

Target: stderr.failure

Meaning

The store reports a version conflict or unavailability.

Source: OUT-BOOKING-CREATE-DENIED

Target: result.out-booking-create-denied

Meaning

The trusted caller context does not authorize this case.

Requirement refs: RF-BOOKING-CREATE-01, RF-BOOKING-CREATE-02, RF-BOOKING-CREATE-03, RF-BOOKING-CREATE-04, RF-BOOKING-CREATE-TITLE, RF-BOOKING-CREATE-AUTH

Resources: none

Examples

```
booking create BOOK001 "Annual review"
```

INT-BOOKING-CANCEL

Name

Cancel command

Description

Accepts a code and presents cancellation, already-cancelled, not-found, or failure.

Illustration only; no channel-specific UI or command choice is promoted to a domain RF by appearing here.

Use case: UC-BOOKING-CANCEL

Trigger

The clerk invokes booking cancel.

Input mapping

Source: argv.code

Target: UC-BOOKING-CANCEL.input.code

Meaning

The booking code argument.

Result mapping

Source: OUT-BOOKING-CANCEL-DONE

Target: stdout.cancelled

Meaning

The booking is cancelled and retained.

Source: OUT-BOOKING-CANCEL-ALREADY

Target: stderr.already

Meaning

The booking is already cancelled.

Source: OUT-BOOKING-CANCEL-NOTFOUND

Target: stderr.not_found

Meaning

The lookup did not find the booking.

Source: OUT-BOOKING-CANCEL-FAILURE

Target: stderr.failure

Meaning

The store or lookup failed.

Source: OUT-BOOKING-CANCEL-INVALID

Target: result.out-booking-cancel-invalid

Meaning

Lookup refused an invalid code.

Source: OUT-BOOKING-CANCEL-DENIED

Target: result.out-booking-cancel-denied

Meaning

The trusted caller context does not authorize this case.

Requirement refs: RF-BOOKING-CANCEL-01, RF-BOOKING-CANCEL-02, RF-BOOKING-CANCEL-03, RF-BOOKING-CANCEL-04, RF-BOOKING-CANCEL-05, RF-BOOKING-CANCEL-AUTH

Resources: none

Examples

```
booking cancel BOOK001
```

INT-BOOKING-LOOKUP

Name

Lookup command

Description

Accepts a code and presents the booking, not found, or failure.

Illustration only; no channel-specific UI or command choice is promoted to a domain RF by appearing here.

Use case: UC-BOOKING-LOOKUP

Trigger

The clerk invokes booking lookup.

Input mapping

Source: argv.code

Target: UC-BOOKING-LOOKUP.input.code

Meaning

The booking code argument.

Result mapping

Source: OUT-BOOKING-LOOKUP-FOUND

Target: stdout.booking

Meaning

The booking is returned.

Source: OUT-BOOKING-LOOKUP-NOTFOUND

Target: stderr.not_found

Meaning

No booking has the code.

Source: OUT-BOOKING-LOOKUP-FAILURE

Target: stderr.failure

Meaning

The store is unavailable.

Source: OUT-BOOKING-LOOKUP-INVALID

Target: result.out-booking-lookup-invalid

Meaning

The booking code is invalid.

Source: OUT-BOOKING-LOOKUP-DENIED

Target: result.out-booking-lookup-denied

Meaning

The trusted caller context does not authorize this case.

Requirement refs: RF-BOOKING-LOOKUP-01, RF-BOOKING-LOOKUP-02, RF-BOOKING-LOOKUP-03, RF-BOOKING-LOOKUP-04, RF-BOOKING-LOOKUP-AUTH

Resources: none

Examples

`booking lookup BOOK001`

Assumptions: none

Sources: SRC-BOOKING-BRIEF

Open issues: none

Appendix: graphs

Graph `GRAPH-VLD-BOOKING-CODE`

Booking code validation

Kind: flow

Nodes

Node `N-VLD-BOOK-CODE-START` (start)

Start

Node `N-VLD-BOOK-CODE-READ` (action)

Read code

Node `N-VLD-BOOK-CODE-EMPTY` (decision)

Non-empty?

Node `N-VLD-BOOK-CODE-LENGTH` (decision)

1 to 12 characters?

Node `N-VLD-BOOK-CODE-CHARS` (decision)

Uppercase letters or digits?

Node `N-VLD-BOOK-CODE-PASS` (result)

Valid

Node `N-VLD-BOOK-CODE-FAIL` (refusal)

Invalid code

Edges

Edge `E-VLD-BOOK-CODE-1`: `N-VLD-BOOK-CODE-START` → `N-VLD-BOOK-CODE-READ`

Requirement refs: none

Edge `E-VLD-BOOK-CODE-2`: `N-VLD-BOOK-CODE-READ` → `N-VLD-BOOK-CODE-EMPTY`

Requirement refs: none

Edge `E-VLD-BOOK-CODE-3`: `N-VLD-BOOK-CODE-EMPTY` → `N-VLD-BOOK-CODE-FAIL`

Label

no

Requirement refs: none

Edge E-VLD-BOOK-CODE-4: N-VLD-BOOK-CODE-EMPTY → N-VLD-BOOK-CODE-LENGTH

Label

yes

Requirement refs: none

Edge E-VLD-BOOK-CODE-5: N-VLD-BOOK-CODE-LENGTH → N-VLD-BOOK-CODE-FAIL

Label

no

Requirement refs: none

Edge E-VLD-BOOK-CODE-6: N-VLD-BOOK-CODE-LENGTH → N-VLD-BOOK-CODE-CHARS

Label

yes

Requirement refs: none

Edge E-VLD-BOOK-CODE-7: N-VLD-BOOK-CODE-CHARS → N-VLD-BOOK-CODE-FAIL

Label

no

Requirement refs: none

Edge E-VLD-BOOK-CODE-8: N-VLD-BOOK-CODE-CHARS → N-VLD-BOOK-CODE-PASS

Label

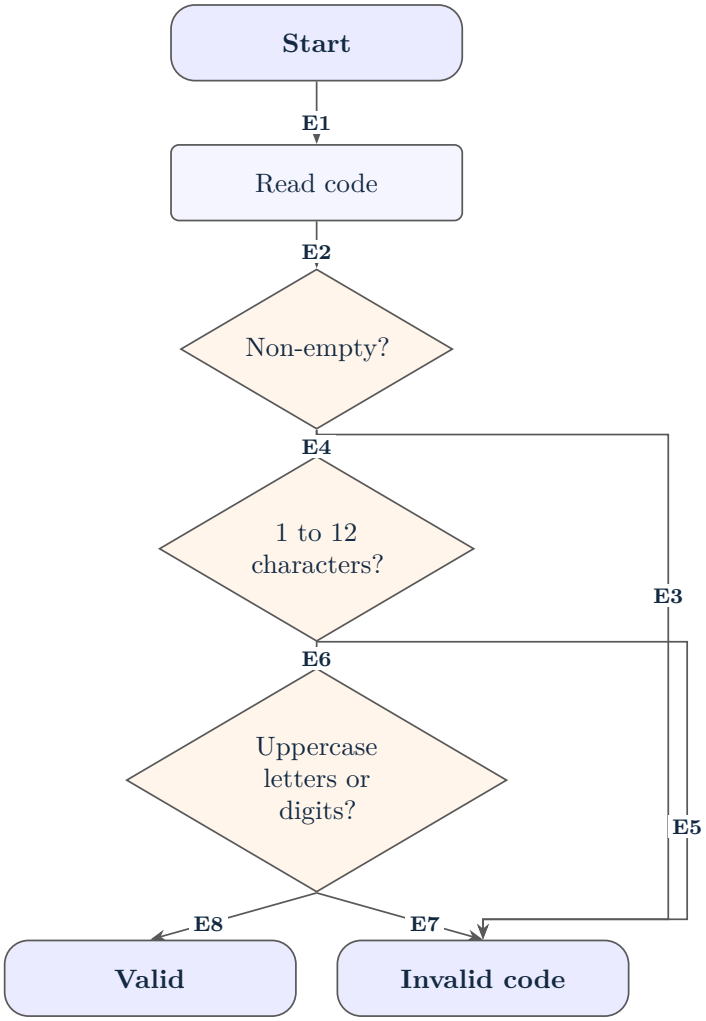
yes

Requirement refs: none
Topology of GRAPH-VLD-BOOKING-CODE:
Nodes:
- N-VLD-BOOK-CODE-START (start)
- N-VLD-BOOK-CODE-READ (action)
- N-VLD-BOOK-CODE-EMPTY (decision)
- N-VLD-BOOK-CODE-LENGTH (decision)
- N-VLD-BOOK-CODE-CHARS (decision)
- N-VLD-BOOK-CODE-PASS (result)
- N-VLD-BOOK-CODE-FAIL (refusal)

Edges:
- E-VLD-BOOK-CODE-1: N-VLD-BOOK-CODE-START -> N-VLD-BOOK-CODE-READ
- E-VLD-BOOK-CODE-2: N-VLD-BOOK-CODE-READ -> N-VLD-BOOK-CODE-EMPTY
- E-VLD-BOOK-CODE-3: N-VLD-BOOK-CODE-EMPTY -> N-VLD-BOOK-CODE-FAIL
- E-VLD-BOOK-CODE-4: N-VLD-BOOK-CODE-EMPTY -> N-VLD-BOOK-CODE-LENGTH
- E-VLD-BOOK-CODE-5: N-VLD-BOOK-CODE-LENGTH -> N-VLD-BOOK-CODE-FAIL
- E-VLD-BOOK-CODE-6: N-VLD-BOOK-CODE-LENGTH -> N-VLD-BOOK-CODE-CHARS
- E-VLD-BOOK-CODE-7: N-VLD-BOOK-CODE-CHARS -> N-VLD-BOOK-CODE-FAIL
- E-VLD-BOOK-CODE-8: N-VLD-BOOK-CODE-CHARS -> N-VLD-BOOK-CODE-PASS

Booking code validation

GRAPH-VLD-BOOKING-CODE



Edge key

E1 E-VLD-BOOK-CODE-1
E2 E-VLD-BOOK-CODE-2
E3 E-VLD-BOOK-CODE-3 — no
E4 E-VLD-BOOK-CODE-4 — yes
E5 E-VLD-BOOK-CODE-5 — no
E6 E-VLD-BOOK-CODE-6 — yes
E7 E-VLD-BOOK-CODE-7 — no
E8 E-VLD-BOOK-CODE-8 — yes

Graph GRAPH-VLD-BOOKING-CANCEL

Cancellation transition validation

Kind: flow

Nodes

Node N-VLD-BOOK-CANCEL-START (start)

Start

Node N-VLD-BOOK-CANCEL-READ (action)

Read booking status

Node N-VLD-BOOK-CANCEL-ACTIVE (decision)

Status active?

Node N-VLD-BOOK-CANCEL-PASS (result)

Transition allowed

Node N-VLD-BOOK-CANCEL-FAIL (refusal)

Already cancelled

Edges

Edge E-VLD-BOOK-CANCEL-1: N-VLD-BOOK-CANCEL-START → N-VLD-BOOK-CANCEL-READ

Requirement refs: none

Edge E-VLD-BOOK-CANCEL-2: N-VLD-BOOK-CANCEL-READ → N-VLD-BOOK-CANCEL-ACTIVE

Requirement refs: none

Edge E-VLD-BOOK-CANCEL-3: N-VLD-BOOK-CANCEL-ACTIVE → N-VLD-BOOK-CANCEL-FAIL

Label

no

Requirement refs: none

Edge E-VLD-BOOK-CANCEL-4: N-VLD-BOOK-CANCEL-ACTIVE → N-VLD-BOOK-CANCEL-PASS

Label

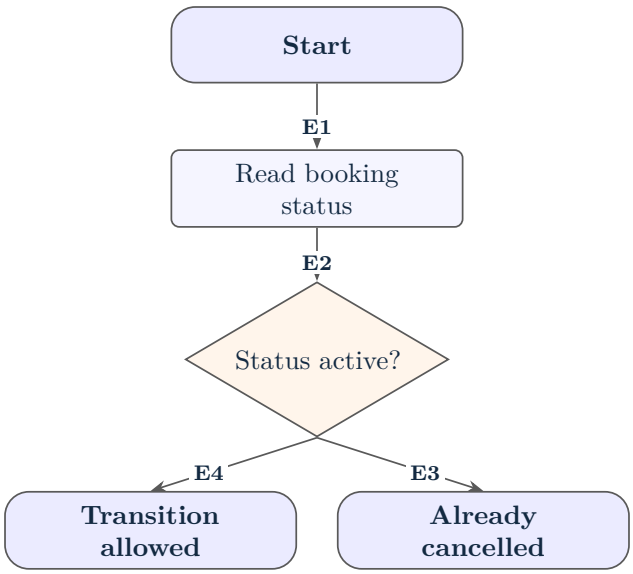
yes

Requirement refs: none

Topology of GRAPH-VLD-BOOKING-CANCEL:
Nodes:
- N-VLD-BOOK-CANCEL-START (start)
- N-VLD-BOOK-CANCEL-READ (action)
- N-VLD-BOOK-CANCEL-ACTIVE (decision)
- N-VLD-BOOK-CANCEL-PASS (result)
- N-VLD-BOOK-CANCEL-FAIL (refusal)
Edges:
- E-VLD-BOOK-CANCEL-1: N-VLD-BOOK-CANCEL-START -> N-VLD-BOOK-CANCEL-READ
- E-VLD-BOOK-CANCEL-2: N-VLD-BOOK-CANCEL-READ -> N-VLD-BOOK-CANCEL-ACTIVE
- E-VLD-BOOK-CANCEL-3: N-VLD-BOOK-CANCEL-ACTIVE -> N-VLD-BOOK-CANCEL-FAIL
- E-VLD-BOOK-CANCEL-4: N-VLD-BOOK-CANCEL-ACTIVE -> N-VLD-BOOK-CANCEL-PASS

Cancellation transition validation

GRAPH-VLD-BOOKING-CANCEL



Edge key

E1 E-VLD-BOOK-CANCEL-1
E2 E-VLD-BOOK-CANCEL-2
E3 E-VLD-BOOK-CANCEL-3 — no
E4 E-VLD-BOOK-CANCEL-4 — yes

Graph GRAPH-BOOKING-LOOKUP

Locate booking

Kind: flow

Nodes

Node N-BOOK-LOOK-START (start)

Start

Node N-BOOKING-LOOKUP-AUTH (decision)

Trusted permission?

Requirement refs: RF-BOOKING-LOOKUP-AUTH

Node N-BOOKING-LOOKUP-DENIED (refusal)

Denied; no collaborator

Requirement refs: RF-BOOKING-LOOKUP-AUTH

Node N-BOOK-LOOK-CODE (validation)

Apply VLD-BOOKING-CODE

Call: validation VLD-BOOKING-CODE

Requirement refs: RF-BOOKING-LOOKUP-01

Node N-BOOK-LOOK-CODEC (decision)

Code valid?

Requirement refs: RF-BOOKING-LOOKUP-01

Node N-BOOK-LOOK-INVALID (refusal)

Refuse: invalid code

Requirement refs: RF-BOOKING-LOOKUP-01

Node N-BOOK-LOOK-LOAD (operation)

Load booking

Call: operation OP-BOOKING-LOAD

Requirement refs: RF-BOOKING-LOOKUP-02

Node N-BOOK-LOOK-NOTFOUND (refusal)

Refuse: not found

Requirement refs: RF-BOOKING-LOOKUP-03

Node N-BOOK-LOOK-FAIL (failure)

Report store unavailable

Requirement refs: RF-BOOKING-LOOKUP-04

Node N-BOOK-LOOK-FOUND (result)

Booking found

Requirement refs: RF-BOOKING-LOOKUP-02

Edges

Edge E-BOOK-LOOK-1: N-BOOK-LOOK-START → N-BOOKING-LOOKUP-AUTH

Requirement refs: none

Edge E-BOOK-LOOK-2: N-BOOK-LOOK-CODE → N-BOOK-LOOK-CODEC

Requirement refs: none

Edge E-BOOK-LOOK-3: N-BOOK-LOOK-CODEC → N-BOOK-LOOK-INVALID

Label

fail

Requirement refs: RF-BOOKING-LOOKUP-01

Edge E-BOOK-LOOK-4: N-BOOK-LOOK-CODEC → N-BOOK-LOOK-LOAD

Label

pass

Requirement refs: RF-BOOKING-LOOKUP-02

Edge E-BOOK-LOOK-5: N-BOOK-LOOK-LOAD → N-BOOK-LOOK-NOTFOUND

Label

NOT_FOUND

Requirement refs: RF-BOOKING-LOOKUP-03

Edge E-BOOK-LOOK-6: N-BOOK-LOOK-LOAD → N-BOOK-LOOK-FAIL

Label

STORE_UNAVAILABLE

Requirement refs: RF-BOOKING-LOOKUP-04

Edge E-BOOK-LOOK-7: N-BOOK-LOOK-LOAD → N-BOOK-LOOK-FOUND

Label

loaded

Requirement refs: RF-BOOKING-LOOKUP-02

Edge E-BOOKING-LOOKUP-ALLOW: N-BOOKING-LOOKUP-AUTH → N-BOOK-LOOK-CODE

Label

authorized

Requirement refs: RF-BOOKING-LOOKUP-AUTH

Edge E-BOOKING-LOOKUP-DENY: N-BOOKING-LOOKUP-AUTH → N-BOOKING-LOOKUP-DENIED

Label

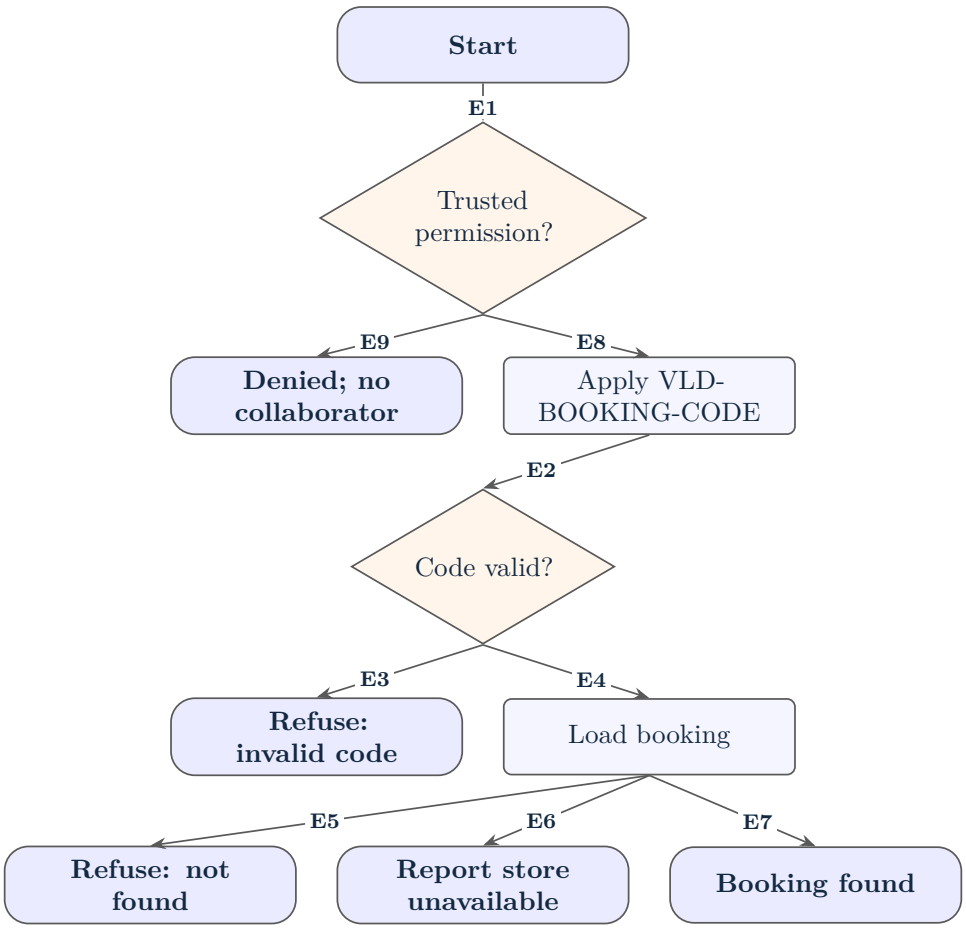
missing / unresolved / insufficient

Requirement refs: RF-BOOKING-LOOKUP-AUTH

Topology of GRAPH-BOOKING-LOOKUP:
Nodes:
- N-BOOK-LOOK-START (start)
- N-BOOKING-LOOKUP-AUTH (decision)
- N-BOOKING-LOOKUP-DENIED (refusal)
- N-BOOK-LOOK-CODE (validation)
- N-BOOK-LOOK-CODEC (decision)
- N-BOOK-LOOK-INVALID (refusal)
- N-BOOK-LOOK-LOAD (operation)
- N-BOOK-LOOK-NOTFOUND (refusal)
- N-BOOK-LOOK-FAIL (failure)
- N-BOOK-LOOK-FOUND (result)
Edges:
- E-BOOK-LOOK-1: N-BOOK-LOOK-START -> N-BOOKING-LOOKUP-AUTH
- E-BOOK-LOOK-2: N-BOOK-LOOK-CODE -> N-BOOK-LOOK-CODEC
- E-BOOK-LOOK-3: N-BOOK-LOOK-CODEC -> N-BOOK-LOOK-INVALID
- E-BOOK-LOOK-4: N-BOOK-LOOK-CODEC -> N-BOOK-LOOK-LOAD
- E-BOOK-LOOK-5: N-BOOK-LOOK-LOAD -> N-BOOK-LOOK-NOTFOUND
- E-BOOK-LOOK-6: N-BOOK-LOOK-LOAD -> N-BOOK-LOOK-FAIL
- E-BOOK-LOOK-7: N-BOOK-LOOK-LOAD -> N-BOOK-LOOK-FOUND
- E-BOOKING-LOOKUP-ALLOW: N-BOOKING-LOOKUP-AUTH -> N-BOOK-LOOK-CODE
- E-BOOKING-LOOKUP-DENY: N-BOOKING-LOOKUP-AUTH -> N-BOOKING-LOOKUP-DENIED

Locate booking

GRAPH-BOOKING-LOOKUP



Edge key

E1 E-BOOK-LOOK-1
E2 E-BOOK-LOOK-2
E3 E-BOOK-LOOK-3 — fail
E4 E-BOOK-LOOK-4 — pass
E5 E-BOOK-LOOK-5 — NOT_FOUND
E6 E-BOOK-LOOK-6 — STORE_UNAVAILABLE
E7 E-BOOK-LOOK-7 — loaded
E8 E-BOOKING-LOOKUP-ALLOW — authorized
E9 E-BOOKING-LOOKUP-DENY — missing / unresolved / insufficient

Graph GRAPH-BOOKING-CREATE

Create booking

Kind: flow

Nodes

Node N-BOOK-CREATE-START (start)

Start

Node N-BOOKING-CREATE-AUTH (decision)

Trusted permission?

Requirement refs: RF-BOOKING-CREATE-AUTH

Node N-BOOKING-CREATE-DENIED (refusal)

Denied; no collaborator

Requirement refs: RF-BOOKING-CREATE-AUTH

Node N-BOOK-CREATE-CODE (validation)

Apply VLD-BOOKING-CODE

Call: validation VLD-BOOKING-CODE

Requirement refs: RF-BOOKING-CREATE-01

Node N-BOOK-CREATE-CODEC (decision)

Code valid?

Requirement refs: RF-BOOKING-CREATE-01

Node N-BOOK-CREATE-INVALID (refusal)

Invalid code or title

Requirement refs: RF-BOOKING-CREATE-01

Node N-BOOK-CREATE-SAVE (operation)

Save active booking

Call: operation OP-BOOKING-SAVE

Requirement refs: RF-BOOKING-CREATE-02

Node N-BOOK-CREATE-DUP (failure)

Report VERSION_CONFLICT

Requirement refs: RF-BOOKING-CREATE-03

Node N-BOOK-CREATE-FAIL (failure)

Report STORE_UNAVAILABLE

Requirement refs: RF-BOOKING-CREATE-03

Node N-BOOK-CREATE-DONE (result)

Booking created

Requirement refs: RF-BOOKING-CREATE-04

Node N-BOOK-CREATE-TITLE (validation)

Apply title validity

Call: validation VLD-BOOKING-TITLE

Requirement refs: RF-BOOKING-CREATE-TITLE

Node N-BOOK-CREATE-TITLEVALID (decision)

Title valid?

Requirement refs: RF-BOOKING-CREATE-TITLE

Edges

Edge E-BOOK-CREATE-1: N-BOOK-CREATE-START → N-BOOKING-CREATE-AUTH

Requirement refs: none

Edge E-BOOK-CREATE-2: N-BOOK-CREATE-CODE → N-BOOK-CREATE-CODEC

Requirement refs: none

Edge E-BOOK-CREATE-3: N-BOOK-CREATE-CODEC → N-BOOK-CREATE-INVALID

Label

fail

Requirement refs: RF-BOOKING-CREATE-01

Edge E-BOOK-CREATE-4: N-BOOK-CREATE-CODEC → N-BOOK-CREATE-TITLE

Label

pass

Requirement refs: RF-BOOKING-CREATE-02

Edge E-BOOK-CREATE-5: N-BOOK-CREATE-SAVE → N-BOOK-CREATE-DUP

Label

VERSION_CONFLICT

Requirement refs: RF-BOOKING-CREATE-03

Edge E-BOOK-CREATE-6: N-BOOK-CREATE-SAVE → N-BOOK-CREATE-FAIL

Label

STORE_UNAVAILABLE

Requirement refs: RF-BOOKING-CREATE-03

Edge E-BOOK-CREATE-7: N-BOOK-CREATE-SAVE → N-BOOK-CREATE-DONE

Label

saved

Requirement refs: RF-BOOKING-CREATE-02, RF-BOOKING-CREATE-04

Edge E-BOOK-CREATE-TITLE-1: N-BOOK-CREATE-TITLE → N-BOOK-CREATE-TITLEVALID

Requirement refs: none

Edge E-BOOK-CREATE-TITLE-2: N-BOOK-CREATE-TITLEVALID → N-BOOK-CREATE-INVALID

Label

fail

Requirement refs: RF-BOOKING-CREATE-TITLE

Edge E-BOOK-CREATE-TITLE-3: N-BOOK-CREATE-TITLEVALID → N-BOOK-CREATE-SAVE

Label

pass

Requirement refs: RF-BOOKING-CREATE-TITLE

Edge E-BOOKING-CREATE-ALLOW: N-BOOKING-CREATE-AUTH → N-BOOK-CREATE-CODE

Label

authorized

Requirement refs: RF-BOOKING-CREATE-AUTH

Edge E-BOOKING-CREATE-DENY: N-BOOKING-CREATE-AUTH → N-BOOKING-CREATE-DENIED

Label

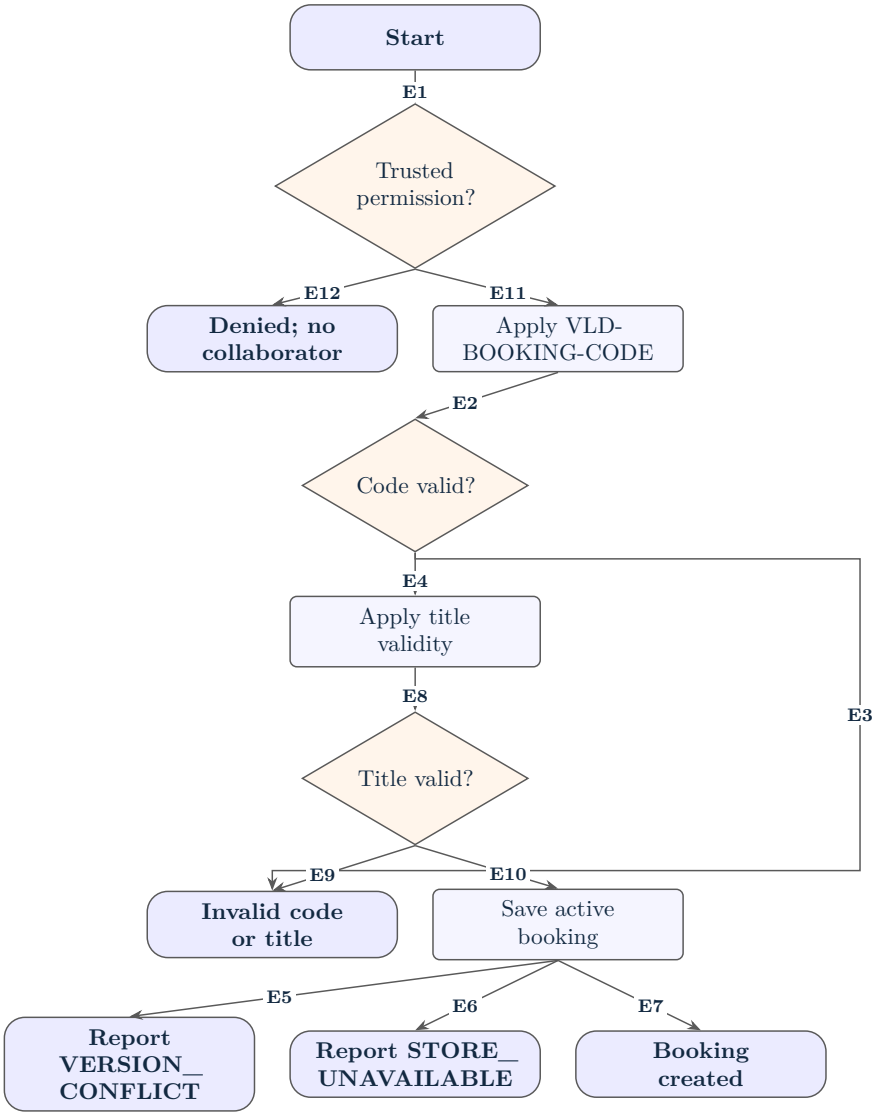
missing / unresolved / insufficient

Requirement refs: RF-BOOKING-CREATE-AUTH
Topology of GRAPH-BOOKING-CREATE:
Nodes:
- N-BOOK-CREATE-START (start)
- N-BOOKING-CREATE-AUTH (decision)
- N-BOOKING-CREATE-DENIED (refusal)
- N-BOOK-CREATE-CODE (validation)
- N-BOOK-CREATE-CODEC (decision)
- N-BOOK-CREATE-INVALID (refusal)
- N-BOOK-CREATE-SAVE (operation)
- N-BOOK-CREATE-DUP (failure)
- N-BOOK-CREATE-FAIL (failure)
- N-BOOK-CREATE-DONE (result)
- N-BOOK-CREATE-TITLE (validation)
- N-BOOK-CREATE-TITLEVALID (decision)

Edges:
- E-BOOK-CREATE-1: N-BOOK-CREATE-START -> N-BOOKING-CREATE-AUTH
- E-BOOK-CREATE-2: N-BOOK-CREATE-CODE -> N-BOOK-CREATE-CODEC
- E-BOOK-CREATE-3: N-BOOK-CREATE-CODEC -> N-BOOK-CREATE-INVALID
- E-BOOK-CREATE-4: N-BOOK-CREATE-CODEC -> N-BOOK-CREATE-TITLE
- E-BOOK-CREATE-5: N-BOOK-CREATE-SAVE -> N-BOOK-CREATE-DUP
- E-BOOK-CREATE-6: N-BOOK-CREATE-SAVE -> N-BOOK-CREATE-FAIL
- E-BOOK-CREATE-7: N-BOOK-CREATE-SAVE -> N-BOOK-CREATE-DONE
- E-BOOK-CREATE-TITLE-1: N-BOOK-CREATE-TITLE -> N-BOOK-CREATE-TITLEVALID
- E-BOOK-CREATE-TITLE-2: N-BOOK-CREATE-TITLEVALID -> N-BOOK-CREATE-INVALID
- E-BOOK-CREATE-TITLE-3: N-BOOK-CREATE-TITLEVALID -> N-BOOK-CREATE-SAVE
- E-BOOKING-CREATE-ALLOW: N-BOOKING-CREATE-AUTH -> N-BOOK-CREATE-CODE
- E-BOOKING-CREATE-DENY: N-BOOKING-CREATE-AUTH -> N-BOOKING-CREATE-DENIED

Create booking

GRAPH-BOOKING-CREATE



Edge key

E1 E-BOOK-CREATE-1
E2 E-BOOK-CREATE-2
E3 E-BOOK-CREATE-3 — fail
E4 E-BOOK-CREATE-4 — pass
E5 E-BOOK-CREATE-5 — VERSION_CONFLICT
E6 E-BOOK-CREATE-6 — STORE_UNAVAILABLE
E7 E-BOOK-CREATE-7 — saved
E8 E-BOOK-CREATE-TITLE-1
E9 E-BOOK-CREATE-TITLE-2 — fail
E10 E-BOOK-CREATE-TITLE-3 — pass
E11 E-BOOKING-CREATE-ALLOW — authorized
E12 E-BOOKING-CREATE-DENY — missing / unresolved / insufficient

Graph GRAPH-BOOKING-CANCEL

Cancel booking

Kind: flow

Nodes

Node N-BOOK-CANCEL-START (start)

Start

Node N-BOOKING-CANCEL-AUTH (decision)

Trusted permission?

Requirement refs: RF-BOOKING-CANCEL-AUTH

Node N-BOOKING-CANCEL-DENIED (refusal)

Denied; no collaborator

Requirement refs: RF-BOOKING-CANCEL-AUTH

Node N-BOOK-CANCEL-LOOK (use_case)

Call UC-BOOKING-LOOKUP

Call: use_case UC-BOOKING-LOOKUP

Requirement refs: RF-BOOKING-CANCEL-01

Node N-BOOK-CANCEL-FOUND (decision)

Booking found?

Requirement refs: RF-BOOKING-CANCEL-01

Node N-BOOK-CANCEL-NOTFOUND (refusal)

Refuse: not found

Requirement refs: RF-BOOKING-CANCEL-01

Node N-BOOK-CANCEL-VALIDATE (validation)

Apply VLD-BOOKING-CANCEL

Call: validation VLD-BOOKING-CANCEL

Requirement refs: RF-BOOKING-CANCEL-02

Node N-BOOK-CANCEL-VALID (decision)

Transition allowed?

Requirement refs: RF-BOOKING-CANCEL-02

Node N-BOOK-CANCEL-ALREADY (refusal)

Refuse: already cancelled

Requirement refs: RF-BOOKING-CANCEL-03

Node N-BOOK-CANCEL-SAVE (operation)

Save cancelled booking

Call: operation OP-BOOKING-SAVE

Requirement refs: RF-BOOKING-CANCEL-04

Node N-BOOK-CANCEL-ERROR (failure)

Report store failure

Requirement refs: RF-BOOKING-CANCEL-05

Node N-BOOK-CANCEL-DONE (result)

Booking cancelled

Requirement refs: RF-BOOKING-CANCEL-04

Node N-BOOK-CANCEL-INVALID (refusal)

Invalid code; no write

Requirement refs: RF-BOOKING-CANCEL-01

Node N-BOOK-CANCEL-LOOKDENIED (refusal)

Lookup denied; no write

Requirement refs: RF-BOOKING-CANCEL-01

Edges

Edge E-BOOK-CANCEL-1: N-BOOK-CANCEL-START → N-BOOKING-CANCEL-AUTH

Requirement refs: none

Edge E-BOOK-CANCEL-2: N-BOOK-CANCEL-LOOK → N-BOOK-CANCEL-FOUND

Label

found / not found

Requirement refs: none

Edge E-BOOK-CANCEL-3: N-BOOK-CANCEL-FOUND → N-BOOK-CANCEL-NOTFOUND

Label

no

Requirement refs: RF-BOOKING-CANCEL-01

Edge E-BOOK-CANCEL-4: N-BOOK-CANCEL-FOUND → N-BOOK-CANCEL-VALIDATE

Label

yes

Requirement refs: RF-BOOKING-CANCEL-01

Edge E-BOOK-CANCEL-5: N-BOOK-CANCEL-VALIDATE → N-BOOK-CANCEL-VALID

Requirement refs: RF-BOOKING-CANCEL-02

Edge E-BOOK-CANCEL-6: N-BOOK-CANCEL-VALID → N-BOOK-CANCEL-ALREADY

Label

fail

Requirement refs: RF-BOOKING-CANCEL-03

Edge E-BOOK-CANCEL-7: N-BOOK-CANCEL-VALID → N-BOOK-CANCEL-SAVE

Label

pass

Requirement refs: RF-BOOKING-CANCEL-04

Edge E-BOOK-CANCEL-8: N-BOOK-CANCEL-SAVE → N-BOOK-CANCEL-ERROR

Label

failure

Requirement refs: RF-BOOKING-CANCEL-05

Edge E-BOOK-CANCEL-9: N-BOOK-CANCEL-SAVE → N-BOOK-CANCEL-DONE

Label

saved

Requirement refs: RF-BOOKING-CANCEL-04

Edge E-BOOK-CANCEL-INVALID: N-BOOK-CANCEL-LOOK → N-BOOK-CANCEL-INVALID

Label

invalid code

Requirement refs: RF-BOOKING-CANCEL-01

Edge E-BOOK-CANCEL-LOOKERROR: N-BOOK-CANCEL-LOOK → N-BOOK-CANCEL-ERROR

Label

lookup failure

Requirement refs: RF-BOOKING-CANCEL-05

Edge E-BOOK-CANCEL-LOOKDENIED: N-BOOK-CANCEL-LOOK → N-BOOK-CANCEL-LOOKDENIED

Label

lookup denied

Requirement refs: RF-BOOKING-CANCEL-01

Edge E-BOOKING-CANCEL-ALLOW: N-BOOKING-CANCEL-AUTH → N-BOOK-CANCEL-LOOK

Label

authorized

Requirement refs: RF-BOOKING-CANCEL-AUTH

Edge E-BOOKING-CANCEL-DENY: N-BOOKING-CANCEL-AUTH → N-BOOKING-CANCEL-DENIED

Label

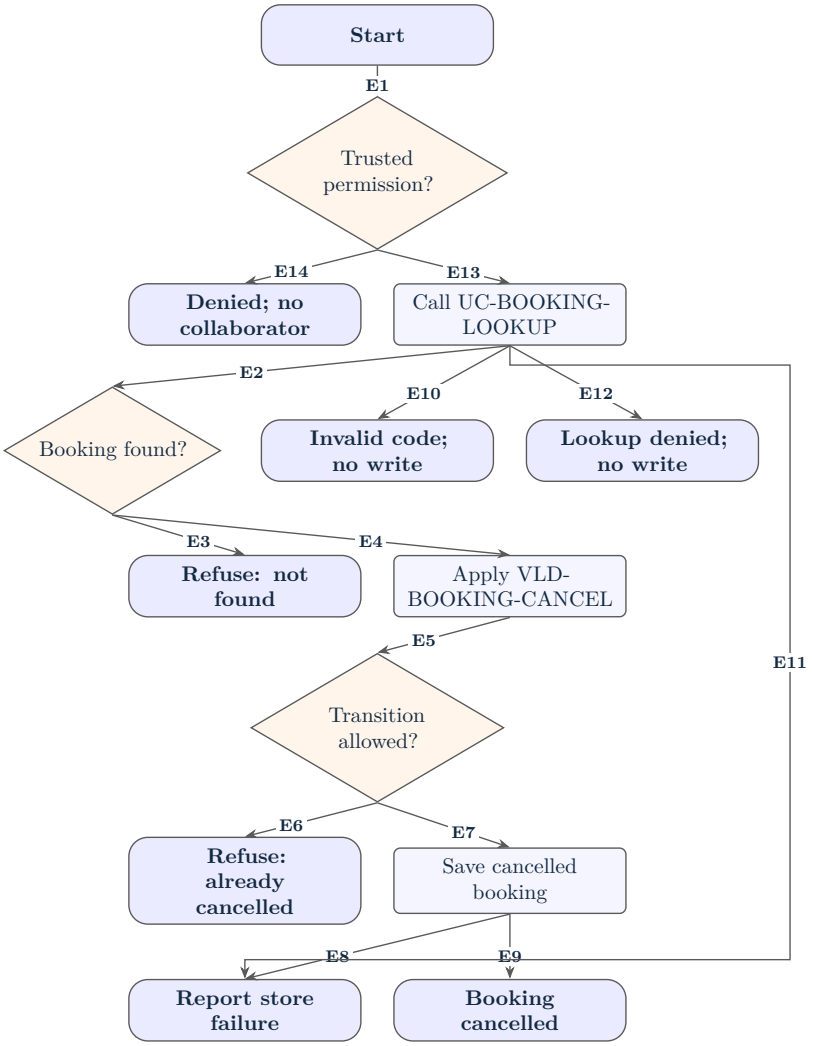
missing / unresolved / insufficient

Requirement refs: RF-BOOKING-CANCEL-AUTH

Topology of GRAPH-BOOKING-CANCEL:
Nodes:
- N-BOOK-CANCEL-START (start)
- N-BOOKING-CANCEL-AUTH (decision)
- N-BOOKING-CANCEL-DENIED (refusal)
- N-BOOK-CANCEL-LOOK (use_case)
- N-BOOK-CANCEL-FOUND (decision)
- N-BOOK-CANCEL-NOTFOUND (refusal)
- N-BOOK-CANCEL-VALIDATE (validation)
- N-BOOK-CANCEL-VALID (decision)
- N-BOOK-CANCEL-ALREADY (refusal)
- N-BOOK-CANCEL-SAVE (operation)
- N-BOOK-CANCEL-ERROR (failure)
- N-BOOK-CANCEL-DONE (result)
- N-BOOK-CANCEL-INVALID (refusal)
- N-BOOK-CANCEL-LOOKDENIED (refusal)
Edges:
- E-BOOK-CANCEL-1: N-BOOK-CANCEL-START -> N-BOOKING-CANCEL-AUTH
- E-BOOK-CANCEL-2: N-BOOK-CANCEL-LOOK -> N-BOOK-CANCEL-FOUND
- E-BOOK-CANCEL-3: N-BOOK-CANCEL-FOUND -> N-BOOK-CANCEL-NOTFOUND
- E-BOOK-CANCEL-4: N-BOOK-CANCEL-FOUND -> N-BOOK-CANCEL-VALIDATE
- E-BOOK-CANCEL-5: N-BOOK-CANCEL-VALIDATE -> N-BOOK-CANCEL-VALID
- E-BOOK-CANCEL-6: N-BOOK-CANCEL-VALID -> N-BOOK-CANCEL-ALREADY
- E-BOOK-CANCEL-7: N-BOOK-CANCEL-VALID -> N-BOOK-CANCEL-SAVE
- E-BOOK-CANCEL-8: N-BOOK-CANCEL-SAVE -> N-BOOK-CANCEL-ERROR
- E-BOOK-CANCEL-9: N-BOOK-CANCEL-SAVE -> N-BOOK-CANCEL-DONE
- E-BOOK-CANCEL-INVALID: N-BOOK-CANCEL-LOOK -> N-BOOK-CANCEL-INVALID
- E-BOOK-CANCEL-LOOKERROR: N-BOOK-CANCEL-LOOK -> N-BOOK-CANCEL-ERROR
- E-BOOK-CANCEL-LOOKDENIED: N-BOOK-CANCEL-LOOK -> N-BOOK-CANCEL-LOOKDENIED
- E-BOOKING-CANCEL-ALLOW: N-BOOKING-CANCEL-AUTH -> N-BOOK-CANCEL-LOOK
- E-BOOKING-CANCEL-DENY: N-BOOKING-CANCEL-AUTH -> N-BOOKING-CANCEL-DENIED

Cancel booking

GRAPH-BOOKING-CANCEL



Edge key

E1 E-BOOK-CANCEL-1
E2 E-BOOK-CANCEL-2 — found / not found
E3 E-BOOK-CANCEL-3 — no
E4 E-BOOK-CANCEL-4 — yes
E5 E-BOOK-CANCEL-5
E6 E-BOOK-CANCEL-6 — fail
E7 E-BOOK-CANCEL-7 — pass
E8 E-BOOK-CANCEL-8 — failure
E9 E-BOOK-CANCEL-9 — saved
E10 E-BOOK-CANCEL-INVALID — invalid code
E11 E-BOOK-CANCEL-LOOKERROR — lookup failure
E12 E-BOOK-CANCEL-LOOKDENIED — lookup denied
E13 E-BOOKING-CANCEL-ALLOW — authorized
E14 E-BOOKING-CANCEL-DENY — missing / unresolved / insufficient

Graph GRAPH-BOOKING-CANCEL-REFUSAL

Cancellation refusal subflow

Kind: flow

Nodes

Node N-BOOK-REFUSE-START (start)

Refusal

Requirement refs: RF-BOOKING-CANCEL-03

Node N-BOOK-REFUSE-ALREADY (refusal)

Already cancelled: no effect

Requirement refs: RF-BOOKING-CANCEL-03

Node N-BOOK-REFUSE-NOTFOUND (refusal)

Not found: no effect

Requirement refs: RF-BOOKING-CANCEL-01

Edges

Edge E-BOOK-REFUSE-1: N-BOOK-REFUSE-START → N-BOOK-REFUSE-ALREADY

Label

already cancelled

Requirement refs: RF-BOOKING-CANCEL-03

Edge E-BOOK-REFUSE-2: N-BOOK-REFUSE-START → N-BOOK-REFUSE-NOTFOUND

Label

not found

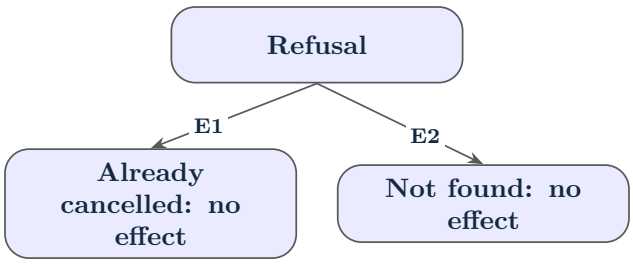
Requirement refs: RF-BOOKING-CANCEL-01

Topology of GRAPH-BOOKING-CANCEL-REFUSAL:
Nodes:
- N-BOOK-REFUSE-START (start)
- N-BOOK-REFUSE-ALREADY (refusal)
- N-BOOK-REFUSE-NOTFOUND (refusal)

Edges:
- E-BOOK-REFUSE-1: N-BOOK-REFUSE-START -> N-BOOK-REFUSE-ALREADY
- E-BOOK-REFUSE-2: N-BOOK-REFUSE-START -> N-BOOK-REFUSE-NOTFOUND

Cancellation refusal subflow

GRAPH-BOOKING-CANCEL-REFUSAL



Edge key

E1 E-BOOK-REFUSE-1 — already cancelled
E2 E-BOOK-REFUSE-2 — not found

Graph GRAPH-VLD-BOOKING-TITLE

Booking title validity

Kind: flow

Nodes

Node N-VLD-BOOK-TITLE-START (start)

Receive title

Node N-VLD-BOOK-TITLE-CHECK (decision)

Meaningful text?

Node N-VLD-BOOK-TITLE-PASS (result)

Pass unchanged

Node N-VLD-BOOK-TITLE-FAIL (refusal)

Invalid title

Edges

Edge E-VLD-BOOK-TITLE-1: N-VLD-BOOK-TITLE-START → N-VLD-BOOK-TITLE-CHECK

Requirement refs: none

Edge E-VLD-BOOK-TITLE-2: N-VLD-BOOK-TITLE-CHECK → N-VLD-BOOK-TITLE-PASS

Label

yes

Requirement refs: none

Edge E-VLD-BOOK-TITLE-3: N-VLD-BOOK-TITLE-CHECK → N-VLD-BOOK-TITLE-FAIL

Label

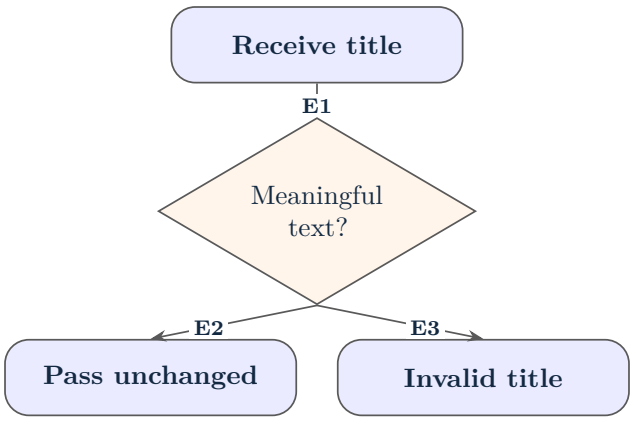
no

Requirement refs: none
Topology of GRAPH-VLD-BOOKING-TITLE:
Nodes:
- N-VLD-BOOK-TITLE-START (start)
- N-VLD-BOOK-TITLE-CHECK (decision)
- N-VLD-BOOK-TITLE-PASS (result)
- N-VLD-BOOK-TITLE-FAIL (refusal)

Edges:
- E-VLD-BOOK-TITLE-1: N-VLD-BOOK-TITLE-START -> N-VLD-BOOK-TITLE-CHECK
- E-VLD-BOOK-TITLE-2: N-VLD-BOOK-TITLE-CHECK -> N-VLD-BOOK-TITLE-PASS
- E-VLD-BOOK-TITLE-3: N-VLD-BOOK-TITLE-CHECK -> N-VLD-BOOK-TITLE-FAIL

Booking title validity

GRAPH-VLD-BOOKING-TITLE



Edge key

E1 E-VLD-BOOK-TITLE-1
E2 E-VLD-BOOK-TITLE-2 — yes
E3 E-VLD-BOOK-TITLE-3 — no

Appendix: sources

SRC-BOOKING-BRIEF

Title

Synthetic booking-desk teaching brief

Uri: null

Revision: 2

Accessed: null

Note

Synthetic example authored for Atlas. No stakeholder interview, independent approval, or accountable human Needs-to-L1 derivation is asserted.

The authorization policy, interaction sketches, and quality targets are didactic choices, not observed business facts.

Appendix: history

HIST-BOOKING-01

Version: 1

Date: 2026-10-25

Description

Initial reference example for the first Atlas Specifications edition, Alpha-v0.1.0. This synthetic example demonstrates the model and its representations; it does not assert stakeholder approval or execution of the described business application.

Derived summary: 5 sections; 1 actors; 1 entities; 1 tools; 3 use cases; 1 boundaries; 7 graphs; 0 resources.